



DynOptic Systems Ltd
www.dynoptic.com

DSL-340 MkIII – Double Pass Particulate Monitor

Measuring in mg/m^3 using DDP

Operators Manual V1.3

DynOptic Systems Ltd – Furlong House, Crowfield, Brackley, Northamptonshire, NN13 5TW United Kingdom

© DynOptic Systems Ltd 2016

All technical details and specifications are subject to change without notice

Contents

Contents.....	1
Product Overview.....	4
Installing the Utility Software and Configuring USB	6
Minimum hardware/software specification for host PC	6
Installing the USB Driver	6
Installing the Utility Software	7
Updating the Utility Software	7
Assigning a valid COM Port.....	7
Using the Utility Software	9
Connecting a PC to the DSL-340	9
The Comms tab.....	10
Connect to Instrument.....	11
Connect Through OI	13
Enable Multidrop.....	13
ID	14
Bus Comms (RS485)	14
The Readings tab	15
Reading	15
Reading Override Mode.....	15
Negative Clipping.....	16
Damping	17
Log Reading - Enabled	17
Log Reading – Append to File	17
Log Reading – Log Q and A	17
Log Reading - Interval.....	18
Graph Button	18
Fault Messages	19
Warning Messages	20
The Outputs tab.....	21
Analogue Output	21
Analogue Low and Analogue High.....	22
4-20mA Output	22
Relays	22
Relay Polarity checkbox.....	22
Threshold.....	23
Delay.....	23
The Settings tab	24
Signal Strength.....	25
AGC button	25
Gain	25
Measure signal	25
Reference signal.....	25

Alignment Tool button	25
Temperature	25
Calibration	26
Path Length	26
DDP Span	26
DDP Zero	26
DDP mg/m ³ Units	26
Enable Opacity	27
Opacity	28
Zero Opacity	28
Calibration Checking	29
Span Check Value	29
Output Function	29
Manual Span Check	29
Automatic Zero and Span Check	30
Interval (Hrs)	30
Enable	30
Zero Drift	31
Span Drift	31
The Diagnostics tab	32
Instrument Information	32
Firmware / PCB Serial Number / Compile Time	32
Backup to PC	33
Restore from PC	33
Restore Factory Defaults	33
Folder Location	34
Direct Comms	34
Using the OI	35
Keypad / Display Fundamentals	35
Moving between Pages	37
Page 0 parameters	38
duSt (Reading)	38
PAgE	38
dAnP (Damping)	38
AnLo & AnHi (Analogue Low & Analogue High)	39
nnA (4-20mA Output)	39
ALr (Threshold)	39
DLy (Delay)	39
rLy (Relay Polarity)	40
driF (Zero Drift & Span Drift)	40
Page 1 parameters	40
duSt (Reading)	41
PAgE	41
dEF (Restore Factory Defaults)	41
Sig (Measure Signal)	41
rEF (Reference Signal)	41
tEnP (Temperature)	42
gAln (Gain)	42
AgC (AGC)	42

PAth (Path).....	42
ZEro (DDP Zero).....	42
SPAn (DDP Span)	42
SAdJ	43
nEg (Negative Clipping).....	43
ccEn (Automatic Zero and Span Check Enabled).....	43
cchr (Calibration Check Hours).....	43
Set-up, Calibration and Commissioning.....	44
Overview	44
Head Alignment	46
Automatic Gain Correction (AGC).....	49
Defining the Path length.....	51
Calibration of Particle Density.....	51
Single Point Calibration (DDP Zero)	52
Single Point Calibration (DDP Span)	52
Multi-Point Calibration (DDP Zero and DDP Span).....	53
Configuring the outputs.....	54
Configuring the Automatic Calibration Check	54
Maintenance	58
Checking the Signal Level	58
Cleaning Optical Surfaces	59
Checking the Reference Signal	59
Fault Messages and Trouble Shooting	60
Fault Messages.....	61
LO REF-VIS ("Lo r").....	61
HI REF-VIS ("Hi r").....	61
HI MEAS-VIS ("Hi S").....	62
OVERRIDE ACTIVE	62
"HEAd" (only displayed on the OI)	62
SPAN CHECK ACTIVE (SP.cc).....	63
ZERO CHECK ACTIVE (Zr.cc).....	63
Warning Messages.....	63
ALARM ACTIVE	63
LO MEAS-VIS ("Lo S").....	63
MEMORY FAULT ("EEch").....	64
COMMS CRC FAULT ("CrC").....	65
Revision Control.....	66

Product Overview

The DSL-340 Double Pass Particulate Monitor measures dynamic fluctuations in open path light transmission, using a transceiver (TRX) and reflector system. It is typically installed on a duct, stack, chimney or flue for the purpose of monitoring increases in particle density (mg/m³) caused by suspended particles (dust and smoke) passing through the light path.

The TRX and Reflector are mounted opposite each other across the stack so that the light beam generated in the TRX passes across the centre of the duct (through the gas stream) and falls directly onto the Reflector on the other side. The Reflector then returns the light to the TRX. Particles moving through the beam will cause fluctuations in the intensity of light received by the TRX.

When calibrated against standard reference measurements, the magnitude of the signal variation relates directly to the dust, smoke and particulate concentration.

A number of checks and measures are employed to maintain continued accuracy of the instrument, including direct monitoring of the light source intensity and the internal instrument temperature of the instrument.

The DSL-340 can be supplied for “stand alone” configuration (i.e. TRX/Reflector heads only – no control unit), in which case command and control of the instrument is performed using either the Utility software provided (for use on a laptop PC), or by ModBus serial comms connected to the TRX. In the stand alone configuration the TRX head has a range of interface outputs including 4-20mA loop, level and service alarm contacts, ModBus, and USB, allowing it to be connected into standard industrial monitoring systems.

Alternatively the DSL-340 can be supplied with an Operator Interface (OI), which is a remote control unit, with 4-digit LED display, 4-button keypad, and full range of interface abilities. The OI is available in an IP65 rated wall mounting enclosure for external use, or a small panel mounting enclosure for installation in a master control panel. The OI has a full range of interface outputs to match those listed for the TRX head above.

Both TRX and Reflector heads are supplied with air-purge bodies, these not only provide the physical mounting point for the instrument, but also allow users to connect a high volume, low pressure air supply to help keep the optics clean and prolong service intervals.

By default the instrument is a 24VDC powered device, but it can be optionally supplied with universal input 90-260VAC PSU for AC operation.

After installing on the stack set-up and the basic calibration involves zeroing the instrument zero under clear path conditions, and performing an upscale calibration adjustment based on comparison between instrument readings and independent gravimetric sample measurements.

The DSL-340 has an automatic zero and span check capability which can be used to check for any performance drift. This automatic check routine can be set to run at a regular, pre-defined time interval using the supplied Utility software.

This manual assumes that the DSL-340 has been properly installed in accordance with the separate installation manual.

This manual covers the set-up, maintenance, and general operation of a DSL-340 monitor using either the DSL-340 Utility Software supplied, and/or the optional OI.

Installing the Utility Software and Configuring USB



Warning: The Utility Software and the USB Driver MUST BE INSTALLED ON THE PC FIRST i.e. before connecting the DSL-340 to a PC. Connecting the PC to a DSL-340 before installing the Utility Software and the USB Driver will cause problems with installation and may even cause the installation to fail.

Minimum hardware/software specification for host PC

The Utility Software supplied with the DSL-340 is not a large or particularly complex program, and therefore has no requirement for a high performance host PC. Any reasonably modern PC/laptop will be suitable. The only “must have” requirements are:

- Windows XP, or newer, operating system
- All recent operating system Service Packs and upgrades (visit the Microsoft website) must have been installed
- At least one free USB slot (assigned to a COM Port between COM1 and COM8 – see later section on Assigning a Valid COM Port).

Installing the USB Driver

The USB Driver software is supplied on a CD that is shipped with the instrument. Alternatively a link to a download file can be provide by DynOptic Systems on request.

1. Ensure that you are logged onto your PC with Administrative rights, so that you have the necessary permission to install software.
2. Copy the USB Driver “CDM xxxxx.exe” file from the CD to any suitable location on the host PC (“My Documents” is a good location).
3. Double click on the .exe file and follow the onscreen installation prompts.

Installing the Utility Software

The DynOptic Systems DSL-340 uses the “DSL-3xx Utility Software” which can be found on the CD that was shipped with the instrument. A link to a download file can be provide by DynOptic Systems on request.

1. Ensure that you are logged onto your PC with Administrative rights, so that you have the necessary permission to install software.
2. Copy the file “DSL-3xx Utility Install xxx.exe” from the CD to any suitable location on the host PC (“My Documents” is a good location).
3. Double click on the .exe file and follow the onscreen installation prompts.

The software should now be installed and should appear in the “All Programs” list in the folder “DSL-3xx UTILITY”. The software is called “DSL-3xx Utility.exe”.

Updating the Utility Software

From time to time DynOptic Systems may release updated versions of the DSL-340 Utility Software. If you receive an updated version, it is essential that you remove any existing version from your PC before installing the new one.

To remove your existing Utility Software use the Windows “Add or Remove Hardware” option from the Windows “Control Panel”.

Assigning a valid COM Port

When you first connect the PC to the DSL-340, it is very likely that the PC will assign the USB connection to a COM Port number outside the acceptable range of COM1 to COM8, in which case the DSL-340 Utility Software will be unable to communicate with the instrument.

You can check which COM Port your PC has assigned to the DSL-340 by opening Windows “Device Manager” before connecting your DSL-340. To open “Device Manager”:

1. Right click on “My Computer”
2. Click on “Manage”
3. Click on “Device Manager”
4. Scroll down to “Ports (COM & LPT)”, then click the “+” symbol to look at the list of COM Ports

Once you can see the list of current COM Ports, connect your PC to a DSL-340 by plugging the USB cable from your PC into the USB connection on top of the TRX head (unscrew the weather-proof cap to gain access).

As you connect, your PC should indicate that it has “Found New Hardware” and a new COM Port number should pop up in the “Ports (COM & LPT)” list.

If the new COM Port is between COM1 and COM8 then your PC has allocated a valid COM Port and no further action is required.

However, if the COM Port number that was assigned to the connection is outside the acceptable range then you must change the assigned COM Port to a number between COM1 and COM8. Follow the procedure below to assign a valid COM Port:

1. Right click on the COM Port (in the “Ports (COM & LPT)” list) that appeared when you connected to the DSL-340
2. Click “Properties”
3. Select the “Port Settings” tab
4. Click on the “Advanced” button
5. Click the “COM Port” and select a new “COM Port Number” between COM1 and COM8 (avoiding any COM Ports which are marked as in use) from the drop down list

Once you have assigned a valid COM Port number to the DSL-340 USB connection your PC will remember this COM Port assignment and you should not need to repeat the process again when connecting your PC to the instrument, provided that you always use the same PC and you always connect via the same USB slot in your computer.

Note; *If you connect to the DSL-340 from a different computer, you will need to repeat the process above and assign a valid COM Port.*

Note; *You should always connect to the DSL-340 using the same USB slot on your PC. If you use a different USB slot you will need to repeat the process above and assign a valid COM Port again, because the PC sees each USB slot as a separate entity.*

Using the Utility Software



Warning: The Utility Software and the USB Driver MUST BE INSTALLED ON THE PC FIRST i.e. before connecting the PC to a DSL-340. Connecting the PC to a DSL-340 before installing the Utility Software and the USB Driver will cause problems with installation and may even cause the installation to fail.

Connecting a PC to the DSL-340

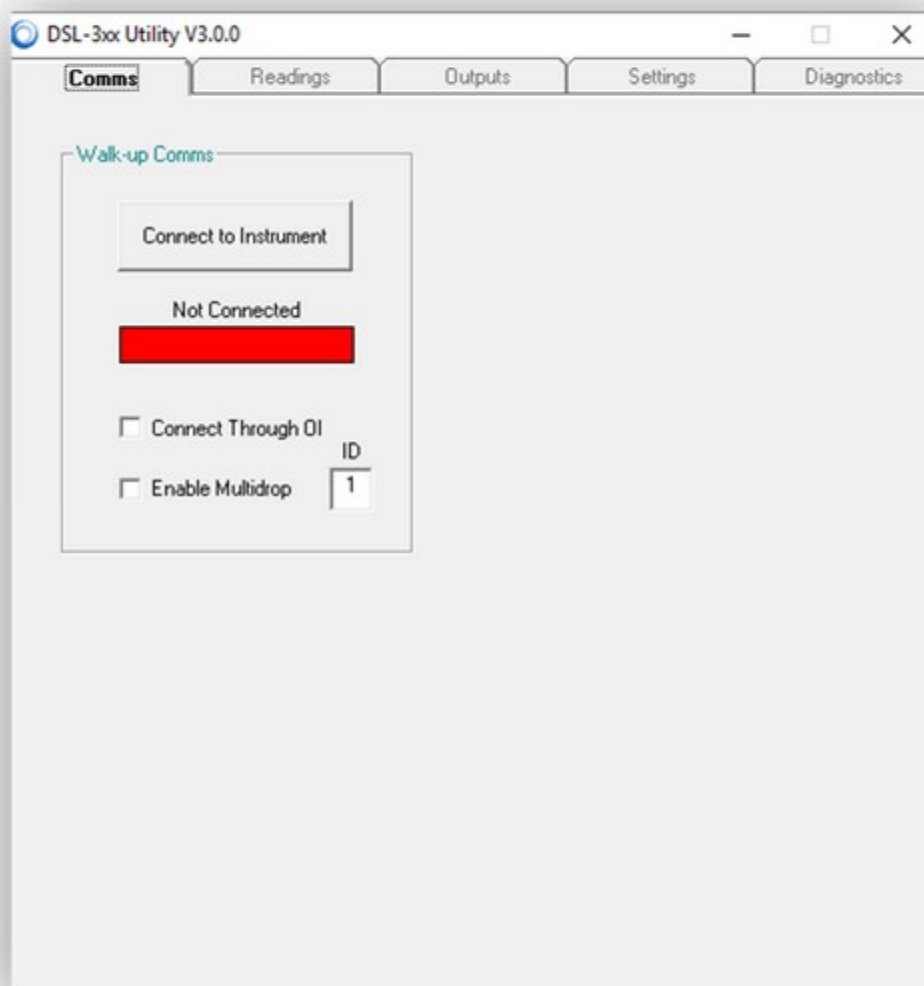
Before connecting the PC to the DSL-340 you **MUST** have installed the USB Driver, and assigned a valid COM Port – see the previous section on installing software.

With the driver installed and com port assigned, connect your PC to the DSL-340 using a standard USB cable (type A to Type B). The USB connection can be found on the side of the TRX head (under a weather-proof cap).

Alternatively it is possible to connect the PC to the optional operator interface (OI). The USB connection for the OI can be found on terminal board inside the terminal compartment. Although this is an indirect connection to the DSL-340, the functionality allowed is exactly the same as the direct connection to the TRX head.

The Comms tab

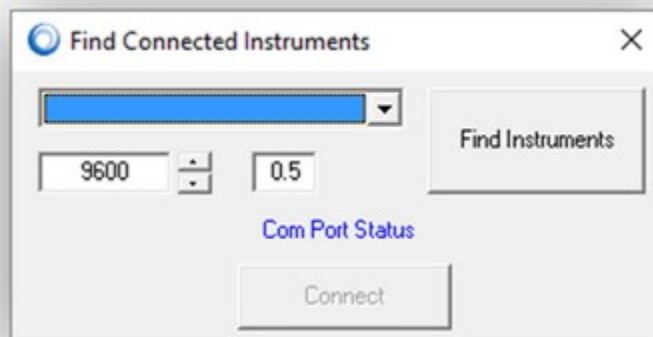
On first running the Utility software you will always arrive at the Comms tab. (see below).



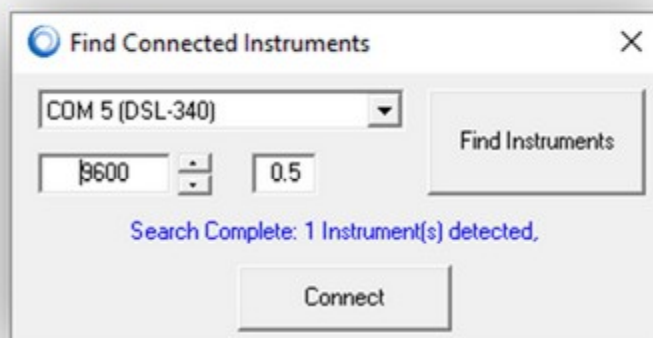
The Comms tab allows you to establish a connection with the DSL-340. When you first run the program there will be no connection, so the status bar will be red (indicating no connection). You will not be able to select any other tab until a comms connection has been established.

Connect to Instrument

To establish a connection with the DSL-340 click the “Connect to Instrument” button. The following pop-up window will appear.

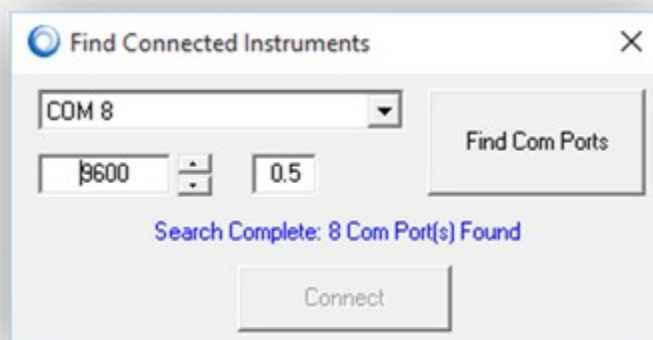
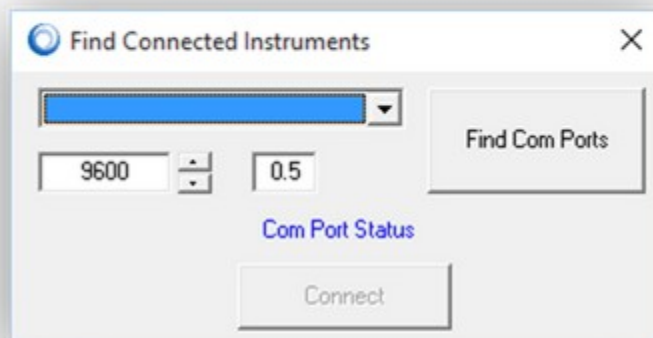


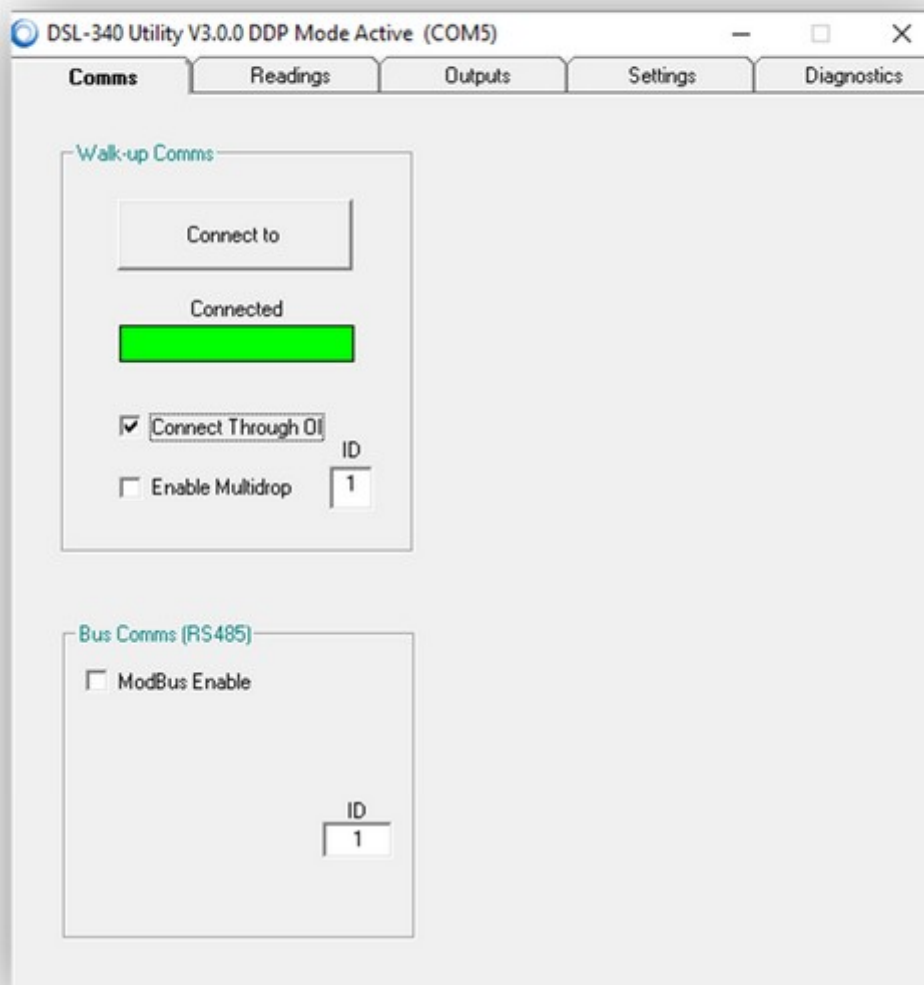
Click the ‘Find Instruments’ button. The software will scan all available computer ports (up to COM 8) for any connected DSL-340 instruments (see below).



Select the required DSL-340 instrument from the list (there will generally only be one) and click ‘Connect’, you will then be returned to the Comms tab where the status indicator bar will now be green (indicating a positive connection).

Note that the function of the ‘Find Instruments’ button can be toggled to ‘Find Com Ports’ by right clicking. Clicking this button will find all of the available Com ports on the PC, as illustrated below.





Once a comms connection is established additional settings become available on the Comms tab and you will be able to select any of the tabs.

Connect Through OI

If you are connected directly to the TRX Head the 'Connect Through OI' checkbox will be un-checked. If you are connected to an OI the 'Connect Through OI' checkbox will be checked. This is an automatic checkbox which you do not need to select or alter.

Enable Multidrop

When connecting to the instrument the Utility will automatically check whether the instrument is set to be in normal or multidrop (multi-heading) mode. For advanced users only, please consult DynOptic Systems if you wish to use multi-heading.

ID

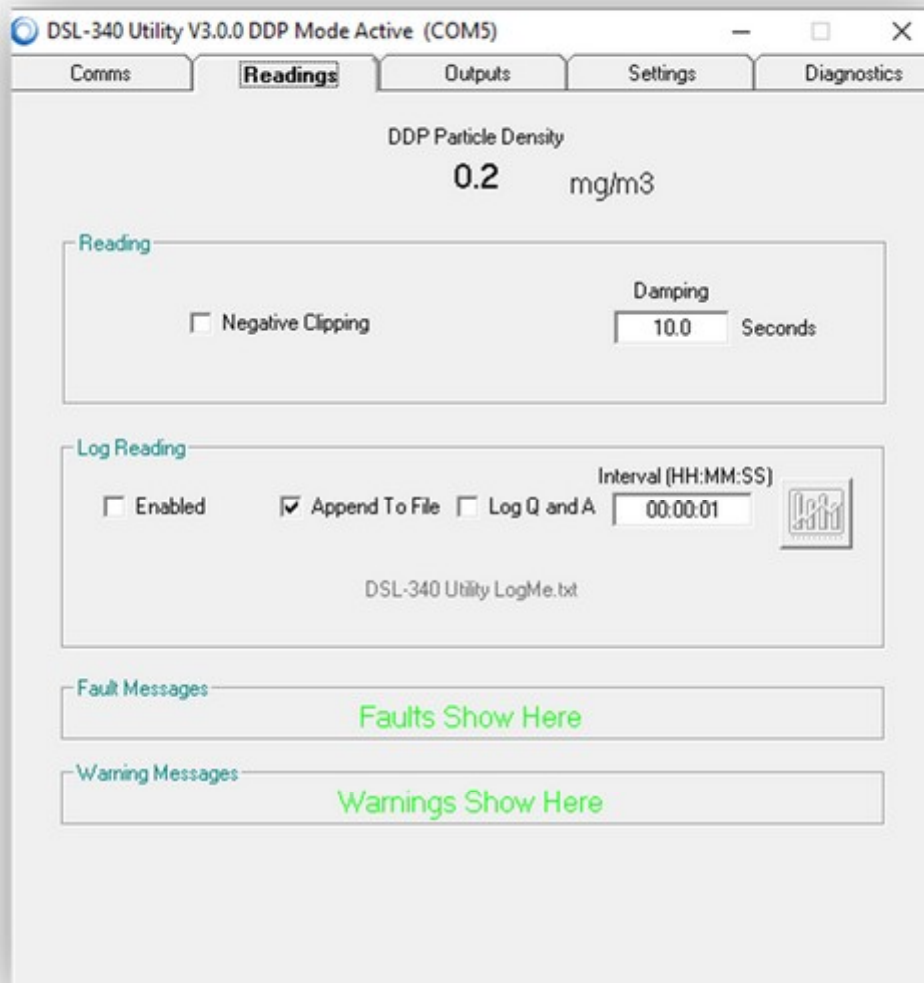
For use when multi-heading on an RS485 bus. The default ID is always "1". For advanced users only, please consult DynOptic Systems if you wish to use multi-heading.

Bus Comms (RS485)

Checking the 'ModBus Enable' tick box changes the RS485 communication to the instrument to be via ModBus. Note that selecting ModBus will prevent any optional OI from communicating with the instrument. For further information please contact DynOptic Systems.

The Readings tab

The Readings tab shows the current measurement reading (complete with units). It allows you to view and alter the damping applied to the measurement reading and to select negative clipping. It also provides checkboxes for activating and controlling automatic data logging (which saves log files directly to the PC hard drive) and shows any current error messages. See the following screen shot.



Reading

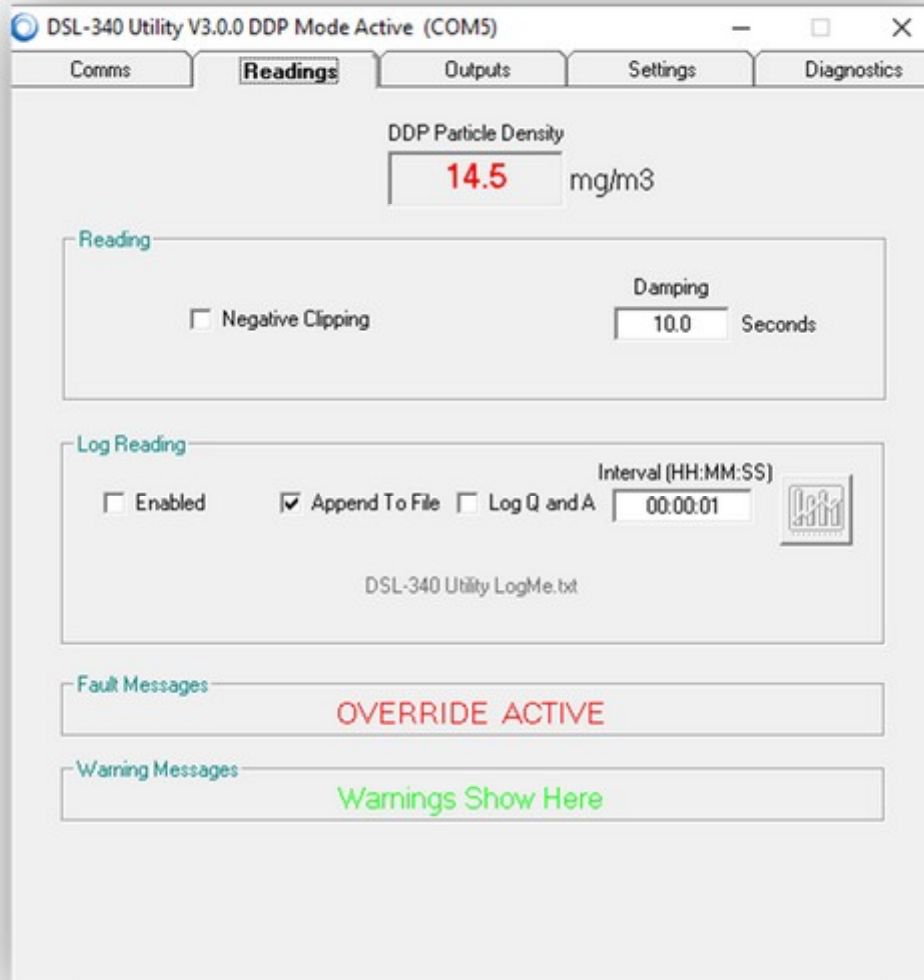
The measurement reading is displayed here. This box is read only.

Reading Override Mode

Double clicking on the measurement reading sets the instrument into the Reading Override Mode.

Note: Entering the Reading Override Mode activates the Data Invalid Alarm.

When in this mode the measurement reading text box is highlighted and the text is red (see below), the user can now manually edit the reading. Typing a value into the measurement reading box and pressing enter forces the reading to the value entered. This function enables you to check if the outputs are setup and are responding correctly at the SCADA or data logging system.



To exit the Reading Override Mode double click in the reading box. Alternatively the instrument will automatically return to normal readings after 5 minutes of inactivity.

Negative Clipping

If this box is UN-CHECKED (default) the instrument will display and output the true reading, even if that reading is negative. Negative readings are possible when the instrument is not set-up and calibrated properly or due to noise fluctuations during operation.

If this box is CHECKED the instrument will clip the measurement reading and the output at zero, so positive readings will show as normal, but when the instrument calculates a negative reading it will hold the display and the outputs at zero (i.e. 0.0mg/m³).

Damping

This value (in seconds) displays the level of Damping applied to the measurement reading. Entering a new value in this box will change the Damping applied. The default Damping for this instrument is 10 seconds. The response time of the instrument to a step change in the reading is approximately three times the damping.

Log Reading - Enabled

When checked, the measured parameter will be logged in a CSV file. The files are stored in a folder in the location 'C:\DynOptic\DSL-340 Utility\READING LOGS', you must be logged onto the PC as administrator for the logs to be stored. The logs will be created at intervals defined by the Interval setting.

The default values logged in the CSV file are the measured reading, the mA output, the instrument status word and the relay status. Contact DynOptic Systems to log other parameters.

Log Reading – Append to File

This box is checked by default. When checked the log file created will be a single file using the electronic serial number of the instrument as a reference in the filename, and each new log is appended within that file. A new file is created each day with the date forming part of the filename.

When the 'Append to file' box is unchecked, the logging will generate a new file at each logging interval, using the time, date and electronic serial number of the DSL-340 as the file name.

The log file(s) can be opened in a text editor or spreadsheet and can be used for data logging, trend graphing or for diagnostics.

Log Reading – Log Q and A

The logging function uses a "question and answer" system to retrieve data for the log. In this system the Utility Software sends a question (an ASCII text string) to the DSL-340, which responds with an answer (usually a number).

When the 'Log Reading' checkbox is un-checked, the log file (in whichever format) only includes the answer, but when this box is checked the log file includes both the question and answer.

By default the question that is asked relates to the measurement reading and general status of the instrument only. However, with advice from DynOptic

Systems, advanced users may modify the question so that the log file records additional or alternative data; for example relay thresholds or analogue output scaling values.

In these circumstances it is beneficial to log both the question and the answer, so that the log file is more easily interpreted.

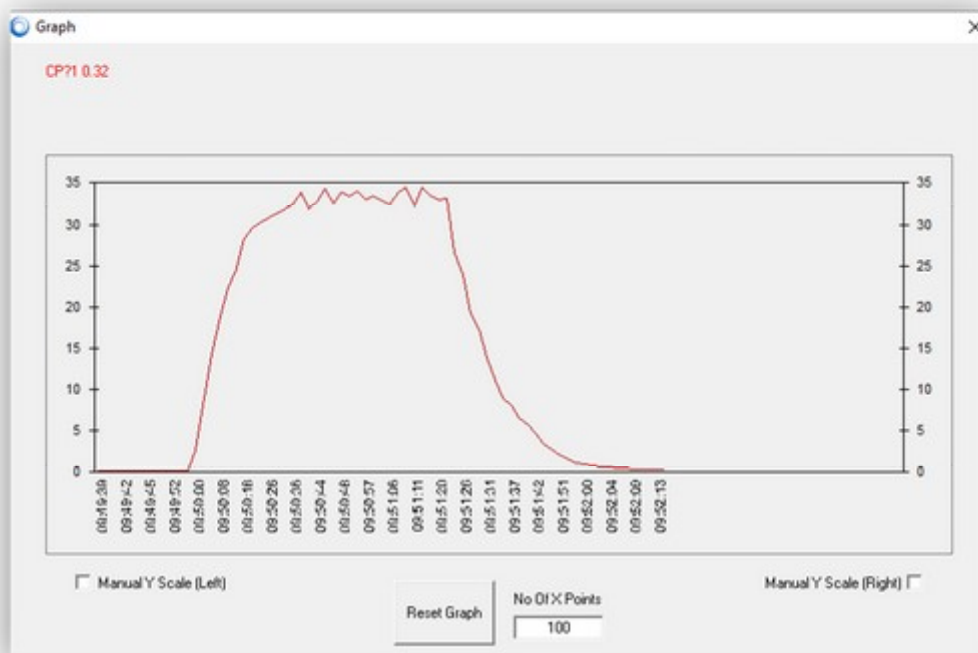
The file that is edited to modify the question is 'DSL-340 UTILITY LogMe.txt'. Left clicking on the filename opens the location folder and allows users to browse the PC to select a different 'LogMe' file. Right clicking on the filename opens the current file in a text editor. Please seek advice from DynOptic Systems before making any changes to this file.

Log Reading - Interval

The time interval between the creation/update of log files is determined here.

Graph Button

As well as a log file being created, the data can be viewed using the graphing feature. While logging click on the 'Graph Icon' a new graph window will appear (see screenshot below).

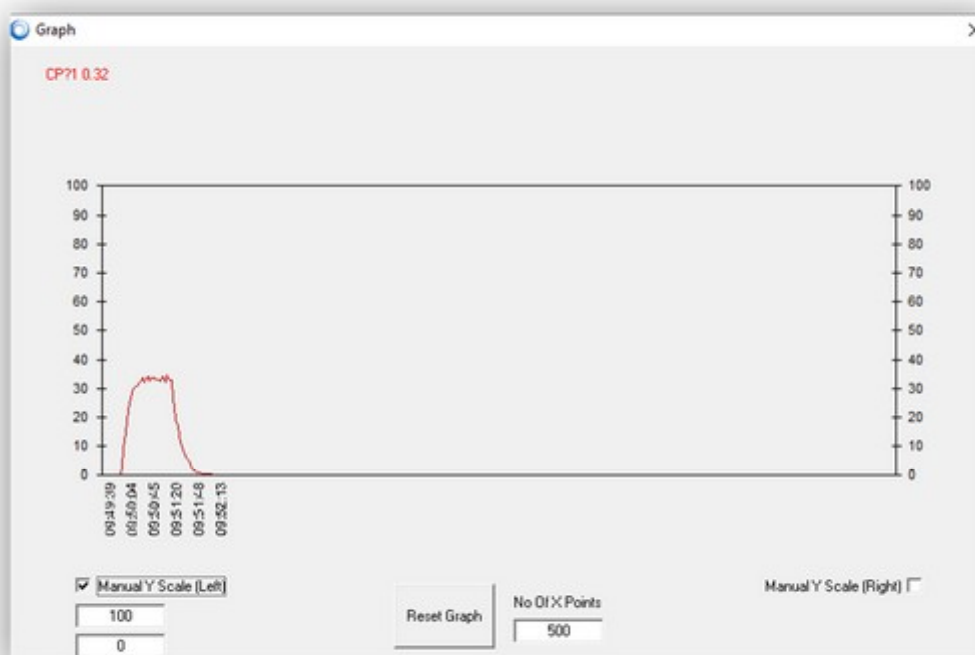


The default file, 'DSL-340 Utility LogMe.txt', only logs the reading (CP?1) and this appears at the top in red 'CP?1'. If the 'DSL-340 Utility LogMe.txt' file has been changed to log several parameters the first four lines of results are displayed on the graph.

There are two 'Y' axis, and by clicking on the coloured text at the top '**CP?1**', the axis that this data is plotted against can be changed; Left = Left Y-Axis, Centre = Not displayed and Right = Right Y-Axis.

The Y-Axis scales are automatic but can be manually changed by checking the 'Manual Y Scale' tick box and then entering the required minimum and maximum values, as illustrated below.

The number of X coordinates can be changed (between 100 and 3600) by entering the value in the 'No Of X Points' box and pressing enter, as illustrated below.



The 'Reset Graph' button clears the currently displayed data.

To exit the graphing feature, close the window using the cross at the top right hand corner of the window to return to the main Utility.

Fault Messages

Under normal operation the message that is displayed is 'Faults Show Here' in green. However, if the instrument detects a fault, the displayed message will switch between 'DATA INVALID' and a brief description of the fault, in red. If there is more than one fault then the messages will switch through each fault description in turn.

Important fault messages will continue to be displayed on this tab until an operator clicks on the message to acknowledge it. If the fault is no longer active the message will disappear, but if the fault status is still active the message will re-appear immediately.

See the section on 'Fault Messages and Troubleshooting' later in this manual for more information.

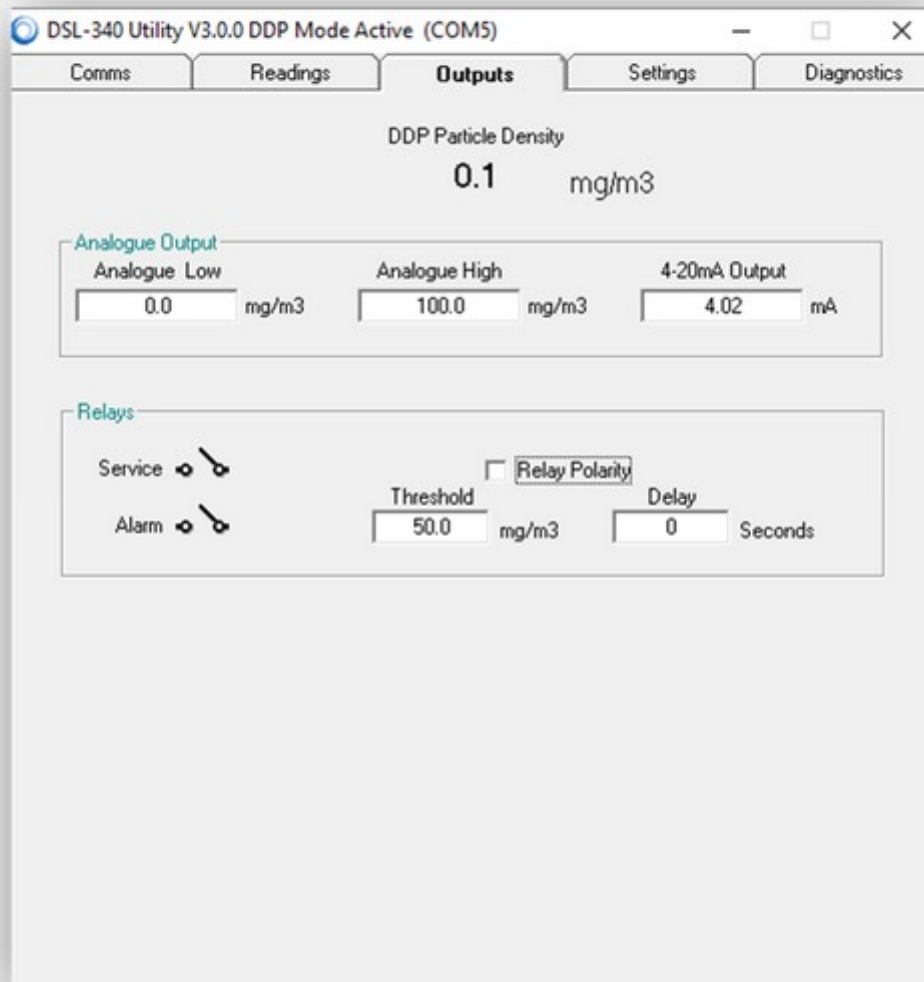
Warning Messages

Under normal operation the message that is displayed is 'Warnings Show Here' in green. If the instrument detects a parameter that is close to or out of range that needs investigating, a brief warning message will be displayed describing the problem, in red. Important warning message will continue to be displayed on this tab until an operator clicks on the message to acknowledge it. If the warning is no longer active the message will disappear, but if the problem causing the warning is still active the message will re-appear immediately.

See the section on 'Fault Messages and Troubleshooting' later in this manual for more information.

The Outputs tab

The Outputs tab allows you to configure electrical interface outputs (such as the analogue output and alarm relay contacts) and also allows you to see the effects of those settings. See the following screen shot.



Analogue Output

The Utility allows the DSL-340 to set-up conditions for it's one (1) analogue output.

Analogue Low and Analogue High

These two parameters set the upper and lower scaling points of the 4-20mA analogue output.

Set the Analogue Low value to the reading at which the analogue output should generate 4mA (usually 0.0mg/m³).

Set the Analogue High value to the reading at which the analogue output should generate 20mA.

4-20mA Output

The 4-20mA Output value is a calculated indication of the expected 4-20mA output; taking into account the Analogue Low and Analogue High settings.

The value shown here is a calculated value only, it is not a measured value from the actual 4-20mA outputs. There is no direct connection between the calculated value and the actual output. However, unless there is a fault with the instrument, the correlation between the calculated value and the real output should be very good.

Relays

The Utility allows the DSL-340 to set-up conditions for the two (2) relays.

The first relay is always the Data Valid (Service) indicator that will go into an alarm condition in the event that the instrument's self-checking identifies a fault.

The second relay is the measurement reading alarm. This operates when the measurement reading exceeds a set Threshold value. The operation of this alarm can be delayed using the Delay setting.

Relay Polarity checkbox

The relay polarity box is UNCHECKED by default.

With the checkbox UNCHECKED, the Data Valid (Service) relay will show an open contact (de-energised) when the data is valid. The relay will be closed (energised) when a fault is detected and the data is invalid. The Alarm relay will be open (de-energised) when the reading is below the threshold value and closed (energised) when the reading exceeds the threshold.

With the checkbox CHECKED, the relay polarity is reversed. The relays will be closed (energised) when there are no faults or the reading is below threshold condition and open (de-energised) when the data is invalid or the measured reading is above the threshold value. This condition represents the recommended failsafe operation.

Threshold

This parameter defines the reading at which the level alarm relay is activated.

Delay

This value determines the continuous length of time (in seconds) that the relay is delayed by.

Delaying the activation of the relay can be used to prevent borderline level changes from “dithering” the relay state. Only genuine, sustained readings in excess of the threshold will actually trigger the alarm.

The Settings tab



Warning: The Settings tab should only be used by a trained instrument engineer who has read and understood the later section on “Setup, Commissioning and Calibration”, as the settings and controls on this tab determine the calibration of the instrument and the measurement reading that it generates.

The Settings tab has a range of settings and controls for initial setup and calibration of the instrument.

DSL-340 Utility V3.0.0 DDP Mode Active (COM5)

Comms Readings Outputs **Settings** Diagnostics

DDP Particle Density
0.1 mg/m³

Signal Strength

AGC Gain 116.129 Measure Signal 4.000 V

Alignment Tool Temperature 40.7 C Reference Signal 4.143 V

Calibration

☐ Enable Opacity DDP mg/m³ Units
☒ mg/m³
☐ mg/Nm³

Path Length 3.0 m

DDP Span 1

DDP Zero 0

Calibration Checking

Span Check Value Medium (133) Manual Span Check Start

Output Function Track

Automatic Zero and Span Check
☐ Enable Interval (Hrs) 24 Zero Drift 0.0 mg/m³
 Disabled Span Drift 0.0 %

Signal Strength

AGC button

Clicking this button will perform an Automatic Gain Correction (AGC), which will adjust the instrument's Gain value to achieve optimum signal strength.

See the later section on Setup, Calibration and Commissioning for further details on using this feature.

Gain

This box displays the Gain value as determined by the AGC routine (above).

This value can be manually changed, but such action should only be undertaken in certain unusual circumstances, and should only be performed by a DynOptic trained engineer, who fully understands the consequences of such action.

See the later section on Set-up, Calibration and Commissioning for further details on using this feature.

Measure signal

This box displays the signal strength from the measurement detector in volts, and is used as a diagnostic tool.

Reference signal

This box displays the signal strength from the reference detector in volts. The reference detector is used to determine changes in the main LED source light output. This value is presented as a diagnostic tool.

Alignment Tool button

Clicking this button will open the Alignment Tool window which will assist in performing final alignment adjustments of the heads.

See the later section on Setup, Calibration and Commissioning for further details on using this feature.

Temperature

This box displays the current circuit board temperature inside the TRX Head and is a diagnostic tool to determine operating conditions.

Calibration

Path Length

The Path Length box is used to define the distance between the TRX and the reflector (the Path length) in metres. The Path length should always be measured “flange-to-flange” i.e. from the face of the flange on the Air-Purge body of the TRX, to the face of the flange on the Air-Purge body of the Reflector. It is necessary to enter this value to accuracy of $\pm 0.05\text{m}$ (5cm).

See the later section on Setup, Calibration and Commissioning for further details on defining this value.

DDP Span

This box displays the current DDP span value which is set to 1.0 by default. This value is manually changed as part of the setup procedure to calibrate the span against a standard reference method.

See the later section on Setup, Calibration and Commissioning for further details on changing this parameter.

DDP Zero

This box displays the current DDP zero value which is set to 0.0 by default. This value can be manually changed as part of the setup procedure to compensate for any clear path offset reading.

See the later section on Setup, Calibration and Commissioning for further details on changing this parameter.

DDP mg/m³ Units

These tick boxes allows the user to select the particle density reading to be displayed as milligrams per meter cubed (mg/m³) or milligrams per normalised meter cubed (mg/Nm³).

If mg/Nm³ is selected some new options appear in the window, as shown below. These allow the user to select the reference or standard gas temperature to be used for the normalisation (either 0°C or 20°C) and input a typical value for the actual gas temperature (in degC) in the stack.

The screenshot shows the 'Calibration' tab of the DSL-340 Utility V3.0.0 DDP Mode Active (COM5) window. It contains the following settings:

- Enable Opacity:** An unchecked checkbox.
- DDP mg/m³ Units:** Two radio buttons, with 'mg/Nm³' selected.
- Normalisation:** Two radio buttons, with 'Std Temp 0 degC' selected.
- Gas Temperature:** A text box containing '0.0' followed by 'degC'.
- Path Length:** A text box containing '3.0' followed by 'm'.
- DDP Span:** A text box containing '1'.
- DDP Zero:** A text box containing '0'.

Enable Opacity

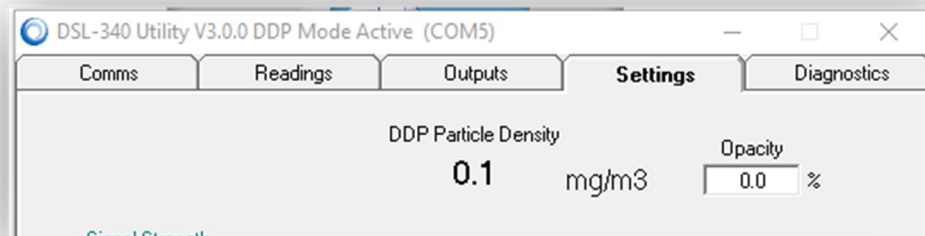
When this tick box is selected the Utility will display the measured opacity as well as the particle density. Selecting the tick box changes the settings tab as shown below.

A box appears on the top row of the settings tab displaying the measured opacity and below the "Enable Opacity" tick box appears a "Zero Opacity" button.

The screenshot shows the 'Settings' tab of the DSL-340 Utility V3.0.0 DDP Mode Active (COM5) window. The 'Enable Opacity' checkbox is now checked, and a 'Zero Opacity' button has appeared below it. The top of the window displays 'DDP Particle Density' as '0.1 mg/m³' and 'Opacity' as '0.0 %'. The 'Signal Strength' section shows 'Gain' at '116.129', 'Measure Signal' at '3.948 V', 'Temperature' at '42.9 C', and 'Reference Signal' at '4.072 V'. The 'Calibration' section at the bottom shows the same settings as the previous screenshot, but with 'mg/m³' selected for units and 'Std Temp 0 degC' for normalisation.

Opacity

This box appears in the top row and displays the measured opacity across the stack. The reading has the same damping as the particle density reading. The measured opacity is only displayed on the settings page of the Utility, it is not available as an analogue output or in the OI. However, it can be read out via the RS485 serial port, contact DynOptic Systems for more information.



Zero Opacity

The “Zero Opacity” button appears underneath “Enable Opacity” tick box. To have a correct measurement of opacity the reading must be set to zero under clear path conditions. When the stack is known to be clear click this button and the opacity reading will be set to 0.0%.



Calibration Checking

This section allows the automatic calibration check to be configured. Note that some of these setup features are only available using the Utility and are not available in the OI menu.

The screenshot shows a 'Calibration Checking' window with three main sections:

- Span Check Value:** A dropdown menu currently set to 'Medium (133)'.
- Output Function:** A dropdown menu currently set to 'Track'.
- Manual Span Check:** A section containing a 'Start' button.
- Automatic Zero and Span Check:** A section containing:
 - An 'Enable' checkbox which is unchecked.
 - An 'Interval (Hrs)' field set to '24'.
 - A 'Zero Drift' field set to '0.1 mg/m3'.
 - A 'Span Drift' field set to '-0.4 %'.
 - The word 'Disabled' is displayed in blue text below the 'Enable' checkbox.

Span Check Value

The instrument produces the automatic span reading by electronically introducing a fixed amount of noise fluctuation onto the measured optical signal. To cover most practical ranges the DSL-340 allows the user to select either a Low, Medium or High setting for the automatic span. The absolute span value associated with each setting is shown in brackets, this value will depend on the path length and DDP Span value.

Once selected this span setting will be used for the automatic or manual span check.

Output Function

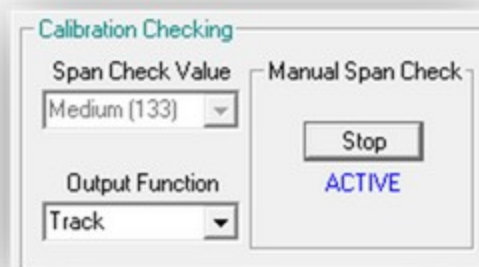
This drop down menu can be used to set the instrument reading to either Track or Hold during a zero or span check.

Track is the default setting, with this is selected the outputs and displays from the instrument will track the readings that occur during the zero and span check.

If Hold is selected then during the zero and span check the instrument output will be held at the last reading before the routine started.

Manual Span Check

Clicking the "Start" button will turn on the internal span function, increasing the particulate density reading by the span setting. When the span is turned on the button changes to "Stop", the word "Active" appears underneath and the instrument will flag a "SPAN CHECK ACTIVE" fault.



To turn off the manual span check click the “Stop” button.

Automatic Zero and Span Check

This section allows the user to set up the automatic zero and span checking. Further details are given in the Set-up, Calibration and Commissioning.

Interval (Hrs)

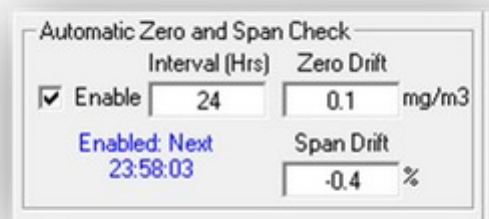
Sets the frequency of the automatic cal checks in hours (1 to 1000), the default setting is 24 hours. Only change this value when the automatic zero and span check is disabled.

Note: the accuracy of the internal clock means that the actual time interval can drift by up to ± 10 minutes / year.

Enable

Selecting the “Enable” tick box starts the automatic zero and span calibration check routine. Please make sure you have set the correct time interval before enabling the calibration check. The first zero and span check will start immediately and takes around 95 seconds to complete.

At the end of the routine a countdown timer appears showing the time until the next automatic zero and span check.



Zero Drift

This displays the DDP zero drift, in units of mg/m³, as measured at the most recent automatic zero check. This measured zero drift will typically be <1mg/m³.

Span Drift

This displays the DDP span drift as a percentage of the span check value, as measured at the most recent automatic span check. The measured span drift will typically be <5%.

The Diagnostics tab

The Diagnostics tab has a number of information only boxes and some features for backup and restore. See the following screen shot. If the Utility software is communicating through an OI then information on the OI firmware and PCB board is also shown.



Instrument Information

There can be two rows of boxes indicating instrument information, the first relates to the DSL-340 instrument itself (the firmware in the TRX circuit board), and the second relates to the OI (the firmware in the main OI circuit board).

Firmware / PCB Serial Number / Compile Time

This shows the serial number and firmware information for the DSL-340 and for the OI (if connected).

Backup to PC

Clicking this button will cause the current instrument settings to be saved in a backup file in the location 'C:\DynOptic\DSL-340 Utility\BACKUP'. The filename of the backup file will incorporate the date, time and electronic serial number of the instrument at time of backup.

It is highly recommended that you use this button to take a backup of your instrument settings after successfully completing your installation, calibration and commissioning. The backup file can be used to restore the instrument's settings, should the live settings ever be lost or become corrupted.

Restore from PC

Clicking this button will allow you to choose a log file from which to restore your instrument settings.

Alarm points and scaling factors will all be overwritten, so it is essential that you have confidence in the log file from which you restore.



Warning: It is highly recommended that you take a backup prior to restoring, so that in the event that you restore from a bad file or unusable settings, you can always restore back to a known good point.



Warning: Restoring from a log file will overwrite ALL settings, parameters and variables, including those that define the zero point and calibration. Restoring from a log file could change the calibration of your instrument.

Restore Factory Defaults

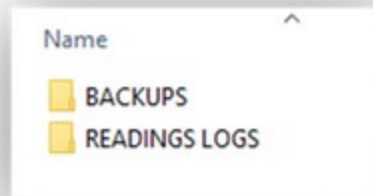
Clicking this button will restore all settings and parameters to default values as determined at our factory.



Warning: This action will overwrite all existing settings and parameters with default values, so your set-up and calibration will be lost, as will any alarm points, scaling factors, and drift values.

Folder Location

This box shows the location on the PC of the folder that contains all of the data stored by the Utility. Left clicking on this filename opens the folder to show three subfolder.



BACKUPS: Is where the instrument backups are stored (see 'Backup to PC' on 'Diagnostics' tab)

READINGS LOGS: Is where the reading logs are stored (see 'Log Reading' on 'Readings' tab)

Direct Comms

Selecting the 'Enable' box opens up an input box, as shown below, which provides direct communication access to the instrument. This can be used to read and edit additional system parameters. This feature should only be used with help from DynOptic Systems.



Using the OI

Keypad / Display Fundamentals

The OI has a 4 digit LED display, four laminated push-buttons, and two LED alarm indicators.



The **Cycle** button cycles you through the available parameters on the Page. It can also be used to cancel data entry.



The **Next** button initiates data entry and moves between digits during data entry.



The **Increase** button is used to increment a digit or move the decimal point during data entry.



The **Enter** button is used to end data entry and store the altered value. It can also be used to re-display a parameters "name".



The level alarm LED will illuminate when the threshold of the level alarm is exceeded.

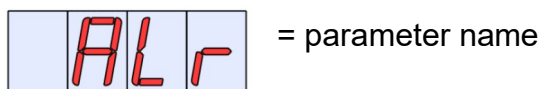


The service alarm LED will illuminate when the instrument detects a fault condition that requires attention.

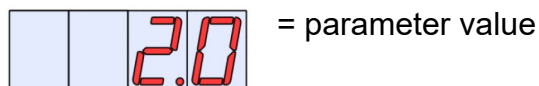
The display will display just one parameter at a time, but the Cycle button can be used to cycle the display through all the available parameters in turn.



When cycling through parameters the display will initially show the "name" of the parameter.



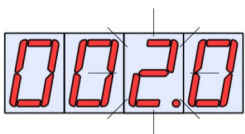
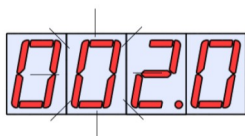
Once cycling is ended, and the display has settled on one parameter for more than one second, the display will switch to show the “value” of that parameter.



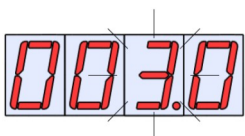
The value of a parameter can be altered by initiating data entry mode, which is achieved by pressing the Next button. On initiating data entry mode the first digit of the value will begin to repeatedly flash indicating that the value of that digit can now be incremented.



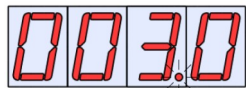
Only one digit can be increased in value at any one time, so select a digit by moving between them with the Next button.



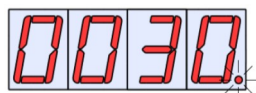
Increase the value of a selected digit using the increase button.



The decimal point can also be selected (if present) by using the Next button also.



The position of the decimal point can be changed using the increase button.



Once the desired value has been achieved in data entry mode, the Enter button can be used to accept that new value and store it in memory. At this point the digits will stop flashing.



It is possible to exit data entry mode without saving an altered value (i.e. to abandon data entry) by pressing the Cycle button before pressing Enter during data entry.

Moving between Pages

The parameters available in the display are arranged on two “pages”.

Page 0 is made up of parameters which are more commonly used, such as output settings, damping, and alarm delays.

Page 1 comprises parameters which are less commonly used or that can be used to alter the calibration or fundamental setup of the instrument.

Switching between Pages is achieved by selecting the Page parameter and toggling between 0 and 1. Because Page 1 contains parameters which can alter the calibration and fundamental setup of the instrument, it is password protected, so upon choosing Page 1 the message PAS? will show briefly, and

the user will then be prompted to enter a 4 digit number (the password) using the usual data entry method.

The default password is 0000.

If the correct number is entered, the message yES will appear and the display will move to Page 1. If the wrong password is entered, the message no will appear and the display will default back to Page 0.

Page 0 parameters

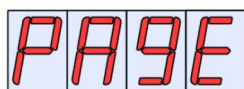
Page 0 contains parameters which are more commonly adjusted and which can only be used to alter the instruments outputs. This page does not contain any parameters which can be used to alter the instruments calibration or fundamental setup, therefore this page is NOT password protected.

The parameters available on Page 0 are listed below by parameter "name" and in the order they appear in the display. In the list below the corresponding Utility Software reference is shown in brackets after the parameter name.



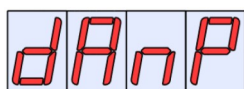
duSt (Reading)

This parameter shows the current measurement reading in mg/m³.



PAGe

This parameter allows you to change between "pages". Choose between 0 and 1 (Page 0 and Page 1). Please note: on choosing Page 1 you will be prompted to enter a password (PAS?) and should enter the password using the normal data entry method. The default password is 0000.



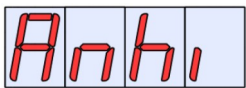
dAnP (Damping)

The value entered here (in seconds) determines the level of damping applied to the reading. The default damping for this instrument is 10 seconds. The response time of the instrument to a step change in the reading is approximately three times the damping.



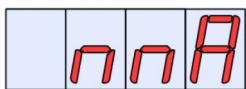
AnLo & AnHi (Analogue Low & Analogue High)

These two parameters set the upper and lower scaling points of the 4-20mA output.



Set the analogue Low value to the reading at which the analogue output should generate 4mA (usually 0.0mg/m³).

Set the analogue high value to the reading at which the analogue output should generate 20mA.



nnA (4-20mA Output)

nnA is the best possible representation of “mA” within the limitations of the LED display. This value is a calculated indication of the expected 4-20mA output; taking into account the Analogue Low and Analogue High settings.

The value shown here is a calculated value, it is not a measured value from the actual 4-20mA outputs. There is no direct connection between the calculated value and the actual output. However, unless there is a fault with the instrument, the correlation between the calculated value and the real output should be very good.

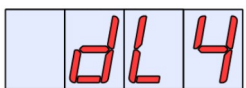


ALr (Threshold)

This value defines the reading at which the level alarm is activated.

When the level alarm is activated it changes the state of the associated alarm relay, and it also lights the level alarm LED on the OI front panel.

The default state of the alarm relay (i.e. energised or de-energised) can be determined with the relay polarity parameter.



DLy (Delay)

This value determines the continuous time (in seconds) that the relay is delayed by.

Delaying the activation of the relay can be used to prevent borderline changes from “dithering” the relay state. Only a genuine sustained reading in excess of the threshold will actually trigger the alarm.



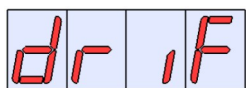
rLy (Relay Polarity)

Changing this value alters the default state of the relays in their non-alarm condition.

With the value set to 0, the relays will be de-energised in a non-alarm condition. In this polarity relays will energise when an alarm is activated.

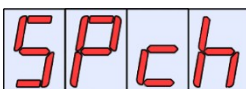
With the value set to 1, the relays will all be energised in a non-alarm condition. In this polarity relays will de-energise when an alarm is activated.

We strongly recommend that users have this value set to 1 so that the relays are used in a “failsafe” configuration.



driF (Zero Drift & Span Drift)

This parameter is only displayed if the instrument has been set to perform automatic zero and span calibration checking. This parameter alternates between displaying the Zero Drift (**Zr.ch**) in units of mg/m³, and the Span Drift (**SP.ch**) as a %.



These values are the drift values measured during the last automatic zero and span check.

Page 1 parameters



Caution; care must be taken

Warning: Page 1 should only be used by a trained instrument engineer who has read and understood the later section on “Setup, Commissioning and Calibration”, as the parameters and functions on this page determine the calibration of the instrument and the measurement reading that it generates.

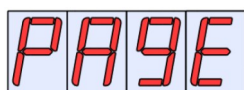
Page 1 has a range of parameters and functions for initial setup and calibration of the instrument, and it is therefore password protected. The default password is 0000.

The parameters available on Page 1 are listed below by parameter “name” and in the order they appear in the display. In the list below the corresponding Utility Software reference is shown in brackets after the parameter name.



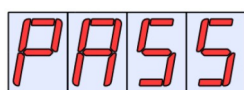
duSt (Reading)

This parameter shows the current measurement reading in mg/m³.



PAgE

This parameter allows you to change between “pages”. Choose between 0 and 1 (Page 0 and Page 1). Please note: on choosing Page 1 you will be prompted to enter a password (PAS?) and should enter the password using the normal data entry method. The default password is 0000.



PASS

This parameter shows the current password. The password can be altered using the normal data entry process.



dEF (Restore Factory Defaults)

This function can be used to restore factory defaults. To use this function the operator must initiate data entry by pressing the Next button, the use then Increase button to select “yES”, and finally press Enter.

Performing a restore Factory Defaults will restore all settings and parameters to default values as determined at our factory.

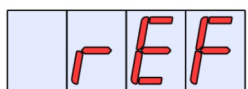


Warning; This action will overwrite all existing settings and parameters with default values, so your complete setup and calibration will be lost, as will any alarm points, scaling factors, damping values etc.



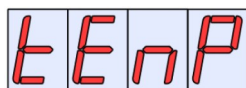
Sig (Measure Signal)

This parameter displays the signal strength on the measurement detector in volts, and is used as a diagnostic tool.



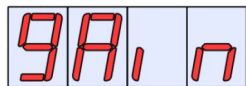
rEF (Reference Signal)

This parameter displays the signal strength on the reference detector in volts. The reference detector is used to determine changes in the main LED source light output. This value is presented as a diagnostic tool.

The image shows a four-digit LED display with the characters 't', 'E', 'n', and 'P' in red. Each character is contained within its own square segment.

tEnP (Temperature)

This parameter displays the current circuit board temperature inside the TRX Head and is a diagnostic tool to determine operating conditions.

The image shows a four-digit LED display with the characters 'g', 'A', 'I', and 'n' in red. Each character is contained within its own square segment.

gAIn (Gain)

This parameter displays the Gain value as determined by the AGC function (below).

This value can be manually changed, but such action should only be undertaken in certain unusual circumstances, and should only be performed by a DynOptic trained engineer, who fully understands the consequences of such action.

The image shows a four-digit LED display with the characters 'A', 'g', and 'C' in red. The first digit segment is empty.

AgC (AGC)

This function enables the operator to perform an Automatic Gain Correction (AGC), which will adjust the instrument's Gain value to achieve optimum signal strengths.

To perform an AGC, first initiate data entry using the Next button, then select "yES" using the Increase button, then select Enter.

The image shows a four-digit LED display with the characters 'P', 'A', 't', and 'H' in red. Each character is contained within its own square segment.

PAth (Path)

The Path parameter is used to define the distance between the TRX and the Reflector (the Path length) in metres. The Path length should always be measured "flange-to-flange" i.e. from the face of the flange on the Reflector Air-Purge body, to the face of the flange on the TRX Air-Purge body.

The image shows a four-digit LED display with the characters '2', 'E', 'r', and '0' in red. Each character is contained within its own square segment.

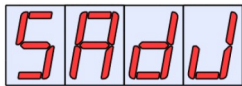
ZErO (DDP Zero)

This parameter displays the current DDP zero value which is set to 0.0 by default. This value can be manually changed as part of the setup procedure to compensate for any clear path offset reading.

The image shows a four-digit LED display with the characters 'S', 'P', 'A', and 'n' in red. Each character is contained within its own square segment.

SPAn (DDP Span)

This parameter displays the current DDP span value which is set to 1.0 by default. This value is manually changed as part of the setup procedure to calibrate the span against a standard reference method.



SAdJ

The DDP span adjust function can be used to force the particulate reading to match the value entered here. This feature should only be performed by a DynOptic trained engineer, who fully understands the consequences of such action.



nEg (Negative Clipping)

If this parameter is set to “no” the instrument will display and output the true reading, even if that reading is negative. Negative readings are possible when the instrument is not set-up and calibrated properly or due to noise fluctuations during operation.

If this parameter is set to “yES” the instrument will clip the measurement reading and the output at zero, so positive readings will show as normal, but when the instrument calculates a negative reading it will hold the display and the outputs at zero (i.e. 0.0mg/m³).



ccEn (Automatic Zero and Span Check Enabled)

This function starts the automatic zero and span calibration check routine. Please make sure you have set the correct DDP Cal Hours before enabling the cal check.

To enable the automatic cal check feature, first initiate data entry using the Next button, then select “yES” using the Increase button, then select Enter

Once the cal check is enabled, the instrument will immediately perform a zero and span check, this takes around 95 seconds to complete.



cchr (Calibration Check Hours)

Sets the frequency of the automatic cal checks in hours (1 to 1000), the default setting is 24 hours. Only change this value when the automatic zero and span check is disabled.

Set-up, Calibration and Commissioning



Warning: *It is essential that all components of the instrument have been properly installed, both physically and electrically, before proceeding with set-up, calibration and commissioning. See the separate DSL-340 Installation Manual for full details of installation requirements.*



Warning: *This manual assumes a basic knowledge of industrial instrumentation and its associated terminology. It is therefore highly recommended that only engineers with instrumentation experiences perform any of the set-up, calibration, and commissioning routines.*



Warning: *when fitting or removing the heads, even for routine maintenance, there is a risk that stack gas may escape from the mounting flange, or air-purge body. Stack gas, can be hot, corrosive or otherwise damaging to health. All necessary precautions must be taken to prevent injury being caused by escaping stack gas.*

Overview

The DSL-340 Particulate Monitor is an in-situ analyser that relies on the local plant structure to determine both the length and angular alignment of its light path (the path along which the light beam travels and along which measurements are made).

The natural variation that can be found between installations, in both length and alignment of the light path, requires the DSL-340 to be capable of working with a wide range of signal strengths (high signal on short paths, low signal on long paths).

This necessity to work with a wide range of signal strengths mean that it is impossible for the DSL-340 to be supplied pre-calibrated or in any way readied for a specific installation.

It is therefore necessary to perform a number of basic set-up, commissioning and calibration routines (after the instrument has been properly installed) to make allowance for the actual signal strengths achieved in any specific

installation. These routines must be performed properly before the instrument can be considered operational or accurate.

Note; *subtle differences between apparently identical installations can produce very different signal strengths, so it is essential that the set-up, calibration and commissioning process is performed independently for each installed instrument.*

Set-up, calibration and commissioning involves performing the following steps, which must be completed in order. The list below is only an overview of the steps required and their order. See later on in this section for a detailed description of how to perform each step:

1. **Perform a Head Alignment** – This optimises the mounting angle of the TRX and Reflector heads to achieve the largest possible measured signal.
2. **Perform an Automatic Gain Correction** – This adjusts the gain applied to the measurement signal to achieve an amplified signal that is within the required voltage range. An AGC should only be performed at instrument setup it must not be repeated after the instrument has been calibrated for particle density.
3. **Define the Path length** – For readings in Particle Density the instrument requires knowledge of the distance (flange-to-flange) between the TRX and the Reflector heads.
4. **Calibration of Particle Density** - The DSL-340 measures the Particle Density in the gas stream in mg/m³, but to do so accurately the instrument must be calibrated to an independent reference measurement (such as isokinetic sampling). This calibration is done by way of the DDP Zero and DDP Span parameters. These are manually adjusted to make the instrument reading match the independent reference measurement.
5. **Configure the outputs** – Once the instrument is calibrated the 4-20mA and Level Alarm outputs can be configured as required for integration with the wider plant systems.
6. **Configure the automatic calibration check** – If automatic zero and span checking is required then this should be configured at the very end of the set-up procedure.

It is absolutely essential that these steps are performed (in order) before considering the instrument operational.

Note; *if at any stage during the operational life of the instrument there is a significant change to the DSL-340 or its installation, it is essential that you repeat the steps above i.e. you must perform the set-up, calibration and*

commissioning procedures all over again. The DDP Span calibration will not need to be repeated but the DDP Zero setting will.

Significant changes include: moving the position or alignment of the heads, changing the path length between the TRX and Reflector, changing out any functional part of the instrument (especially the circuit boards or heads) and changing the Gain value.

Head Alignment

The DSL-340 TRX produces a narrow optical beam that is approximately 3 degrees wide. To achieve a measurable signal this beam must overlap with the aperture of the Reflector head. This means that the angular alignment between the mounting flanges attached to the duct must be less than ± 2 degrees, as defined in the installation manual.

For optimum performance of the DSL-340 the final alignment between the heads is achieved by adjusting the mounting angle of the heads whilst monitoring the measured signal level, which can be viewed using the Utility software or the OI.

To enable this adjustment, the DSL-340 Air-Purge Bodies must be fixed to the mounting flanges with the rubber flange gaskets (supplied) in between, as shown in Figure 1. The angle of the head can then be adjusted by tightening the fixing nuts and compressing the flange gasket in an uneven manner.

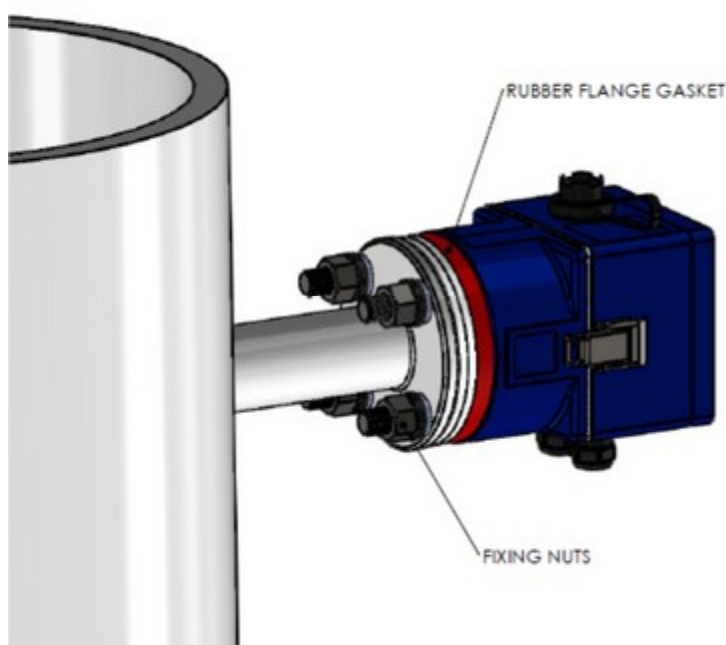


Figure 1: Air-Purge Body mounted to flange with the rubber flange gasket

Note; *the nuts used to compress the flange gasket should not be over tightened or too loose. The recommended torque range is 10 to 50Nm. The flange gasket provides only a small level of adjustment and cannot correct for angular errors between the flanges of more than the specified ± 2 degrees.*

The process of adjustment is to set all four nuts to the minimum torque (10Nm) then individually tighten each one to compress the gasket and hence change the angle of the mounting. When the final alignment has been reached always check that all four nuts are sufficiently tight (at least 10Nm), since strongly adjusting one nut can result in others becoming loose.

The following steps outline the recommended procedure for completing the alignment of the heads. This process assumes the use of a Laser Alignment Tool that is available as an optional extra from DynOptic Systems.

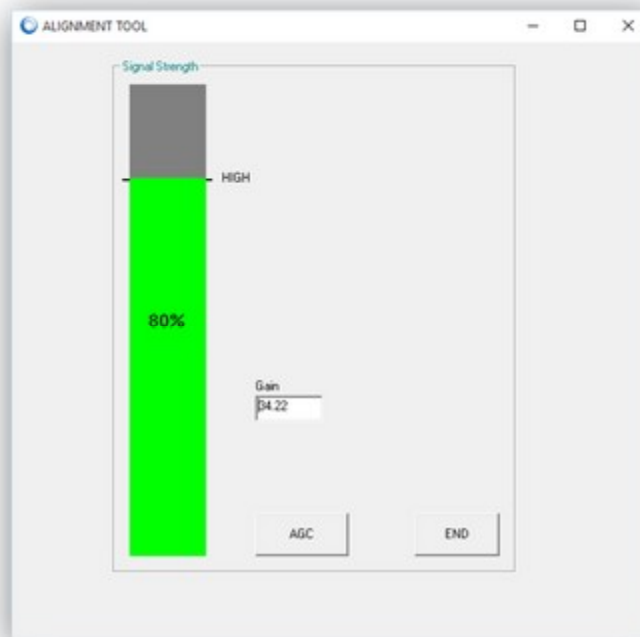
1. Attach the Laser Alignment Tool to the Reflector air-purge body and turn it on. Remove the back of the tool and view through the window of the instrument to see the red laser spot on the far side of the stack. The angle of the Reflector purge head can now be adjusted, using the mounting nuts, to approximately align the laser spot to the centre of the TRX flange on the far side of the stack.

Remove the laser alignment tool and attach the Reflector head to its air-purge body.

2. Attach the Laser Alignment Tool to the TRX air-purge body on the other side of the stack. Repeat the alignment process to align the laser spot to the centre of the Reflector flange on the far side of the stack.

Remove the laser alignment tool and attach the TRX head to its air-purge body.

The two heads are now well aligned but to get the optimum alignment it is necessary to monitor and maximise the measured signal. Power the instrument and use the Utility software to monitor its performance. In the Settings tab select the Alignment Tool button. This will bring up the Alignment Tool window which allows the operator to monitor the measured signal strength, see the following screenshot. Alternatively the measured signal strength can be monitored in the OI using the Sig parameter.



Note: The maximum signal can be more than 80%. Always perform an AGC and re-align for maximum signal.

The alignment tool window has a signal strength bar which increases in height with increases in the measured signal. If the measured signal goes too high (or too low) for the gain range currently set, the colour of the bar turns from green to red. When the bar turns red, click on the AGC button to adjust the instrument Gain to return the bar to green.

Note: The measured signal is slightly damped so it takes a few seconds for the signal strength bar to respond to an adjustment in the head alignment.

3. Adjust the angle of the TRX head by using the mounting nuts to maximise the displayed signal level i.e. to maximise the height of the bar. If the bar goes too high and turns red, stop adjusting and click the AGC button to change the Gain setting, then continue adjusting the angle of the head. Stop adjusting the angle of the head when the maximum signal strength has been achieved with minimum Gain. Ensure that all four mounting nuts are tight (at least 10Nm).
4. There is no need to adjust the angle of the Reflector head. The Reflector head uses corner cube reflectors whose performance is very insensitive to angle.

It is recommended that the instrument is left powered for at least a further 20minutes to stabilise and warm-up before proceeding with the next stage.

Automatic Gain Correction (AGC)

An AGC alters the instrument's gain value to achieve the optimum signal strength of approximately 4V. This must be performed under clear path conditions, defined as follows:

1. The instrument must be properly and permanently installed as described in the separate Installation Manual and the heads must be properly aligned as described above.
2. The instrument must have been powered up and left to settle for a minimum of 1 hour.
3. The optical surfaces of the instrument must be perfectly clean and free from dirt, dust, moisture, or oil.
4. The optical path between the TRX and the Reflector must be clear and free from any obstruction.
5. The duct or stack must be free from any dust, smoke, soot, steam, or particulate (i.e. the process behind the stack must be shut down).

Once clear path conditions have been achieved the gain correction should be performed, either by clicking the AGC button on the Settings tab in the Utility Software, or by selecting the AGC parameter in the OI and toggling to yES.

This will cause the instrument to automatically calculate and store a new gain value to achieve the optimum signal strength.

Note; *the DSL-340 should not be operated with a gain of more than 2000. The instrument is capable of setting higher gains, but these are for use in alignment and setup only. If the AGC produces a gain higher than 2000 then either the path length being used is too long for the particular instrument or the head optics are dirty or they have not been aligned correctly. Alternatively there may be an instrument fault which must be reported to a DynOptic trained engineer or your local distributor.*

The achieved signal strength (Measure Signal) and gain setting can be seen in the Settings page of the Utility software. See the following screen shot.

DSL-340 Utility V3.0.0 DDP Mode Active (COM5)

Comms Readings Outputs **Settings** Diagnostics

DDP Particle Density
0.1 mg/m³

Signal Strength

AGC Gain 116.129 Measure Signal 4.000 V

Alignment Tool Temperature 40.7 C Reference Signal 4.143 V

Calibration

☐ Enable Opacity DDP mg/m³ Units
☒ mg/m³
☐ mg/Nm³

Path Length 3.0 m

DDP Span 1

DDP Zero 0

Calibration Checking

Span Check Value Medium (133) Manual Span Check Start

Output Function Track

Automatic Zero and Span Check
☐ Enable Interval (Hrs) 24 Zero Drift 0.0 mg/m³
 Disabled Span Drift 0.0 %

The level of gain can also be set manually (by overwriting the value in the gain box) but this is NOT recommended.

Sometimes manual Gain corrections are used where the plant cannot be shut down and proper clear path conditions cannot be established. In this instance it is essential that you have an independent knowledge of the actual dust levels in the duct, so that a Gain value can be calculated to give a Measure Signal of approximately 4.0V when the plant is eventually stopped and a clear path is established.



Warning; Setting a Measure Signal of ~4.0V (using either automatic or manual Gain correction) without clear path conditions can cause the Measure Signal to become saturated when the plant stops and a clear path is established. This is because as the light path clears, the Measure Signal will increase. The maximum possible Measure Signal is 5.0V, but any voltage above 4.9V is considered unsuitable for normal operation.

Manual Gain corrections are usually very imprecise (little better than a guess) and should only ever be considered as a temporary solution in non-critical situations. They should be replaced by a proper AGC at the earliest possible opportunity. Please remember that manual Gain corrections are NOT recommended, and should only be performed by engineers with a full understanding of the consequences.



Warning; Changing the Gain value (either automatically or manually) will invalidate any existing calibration. After a Gain correction of any kind, you must always go on to repeat the DDP Zero calibration.

Defining the Path length

Measurements for Particle Density require knowledge of the Path length for use in the calculation.

Path length is defined as the distance in metres between the face of the flange on the Air-Purge body of the TRX, and the same point on the Air-Purge body of the Reflector. It is necessary to enter this value to accuracy of $\pm 0.05\text{m}$ (5cm).

Calibration of Particle Density

The relationship between the measured DDP signal and Particle Density is approximately linear but the slope (DDP Span) and intercept (DDP Zero) will vary for each installation, based on the type, shape, size and colour of the particles in the gas stream.

The DDP Span and DDP Zero values cannot be pre-determined with factory settings and must instead be determined on site. The DDP Zero can be set by measuring a clear path. The DDP Span requires comparison of the instrument readings with an independent reference measurement. Making this comparison and adjusting the DDP Span value accordingly will complete the calibration for the instrument

There are two basic calibration approaches, the simplest is to measure a clear path and a single upscale reading and use these to separately determine the DDP Zero and DDP Span values. The second is to perform several standard measurements over a range of particle density levels in the stack, find a best linear fit, and use this to define the optimum DDP Zero and DDP Span settings.

Single Point Calibration (DDP Zero)

Under clear path conditions the instrument should be reading 0.0mg/m³. However, external noise sources can, under some circumstances, introduce a small zero offset signal. To remove the effect of external influences on the signal it is advisable to perform a clear path check. Under clear path conditions (as defined previously) leave the instrument measuring for at least 10 minutes. If after this time the reading is significantly different from zero (e.g. more than 1mg/m³) then it is advisable to add a zero offset term.

Using the settings page of the Utility software (or the OI) manually input a negative DDP Zero value to correct for the measured clear path offset.

Single Point Calibration (DDP Span)

A standard reference measurement must be taken at the same time, and from a place in the stack very close to the DSL-340 being calibrated, so that the two measurements can be considered coincident.

When taking the coincident measurements the plant must be operational and producing a meaningful concentration of particulates in the gas stream at that instant, ideally close to the maximum expected value.

It is preferable to make coincident measurements when the particle concentration in the stack is at its highest possible, since this improves the accuracy of the calibration. Calibrations performed with reference measurements close to 0mg/m³ are not valid. Note that the DSL-340 is only suitable for measuring Particle Density on a dry stack, if there is a significant quantity of condensed water in the gas flow the relationship between the measured DDP level and reference particle density is no longer valid.

Before proceeding with the calibration measurement it is essential that the instrument has had a clear path zero check and (if required) the DDP Zero set, as described in the previous section.

In order to perform a Particle Density calibration the following conditions for the DSL-340 must be met:

1. The instrument must have been properly and permanently installed, as described in the separate Installation Manual, and the heads aligned for optimum signal.

2. The correct flange-to-flange Path length has been entered in the Path box on the Settings tab in the Utility software.
3. The instrument must have been properly and recently setup (as described earlier in this section of the manual).
4. The instrument has been powered up and left to settle for a minimum of 1 hour.
5. The plant is running and the particulate concentration in the stack is at a high level.

The DSL-340 will show a Particle Density measurement in units of mg/m³, (or mg/Nm³). This reading can be scaled by changing the DDP Span value displayed in the box on the Settings tab of the Utility software (the default DDP Span value is 1.0).

The sampling probe for the reference measurement must be inserted into the duct close to, but not actually in, the optical path of the DSL-340, in this way it will be sampling a very similar gas stream.

During the time period over which the reference sample is taken (typically 30mins) the average Particle Density displayed by the DSL-340 instrument should be recorded (PD_I).

The particle density measured by the standard reference method (PD_{REF}) is then compared to that displayed by the DSL-340 (PD_I). A new DDP Span value is calculated to correct for any differences, using the expression below. This new value should then be entered into the DDP Span box in the Utility software.

$$SPAN = SPAN_0 \left(\frac{PD_{REF}}{PD_I} \right)$$

Where:

SPAN = the new DDP Span value

SPAN₀ = the original DDP Span Value

PD_{REF} = the Particle Density (mg/m³) from the standard reference method

PD_I = the Particle Density (mg/m³) from the DSL-340 instrument

Multi-Point Calibration (DDP Zero and DDP Span)

A more precise linear calibration can be achieved by performing the previously described coincident measurement over a range of different particle densities, from very low to close to maximum. The usual requirement is at least 5 different particle density levels, approximately equally distributed across the required measurement range. To simplify the analysis the default instrument settings should be used during the calibration (i.e. DDP Zero=0.0 and DDP Span= 1.0).

From the measured set of coincident readings a best fit linear relationship can be derived between the instrument readings and the result of the standard reference measurement. This linear fit will produce the new DDP Zero and DDP Span values that can be entered into the instrument using the Settings Page of the UTILITY. An example of this is illustrated below.

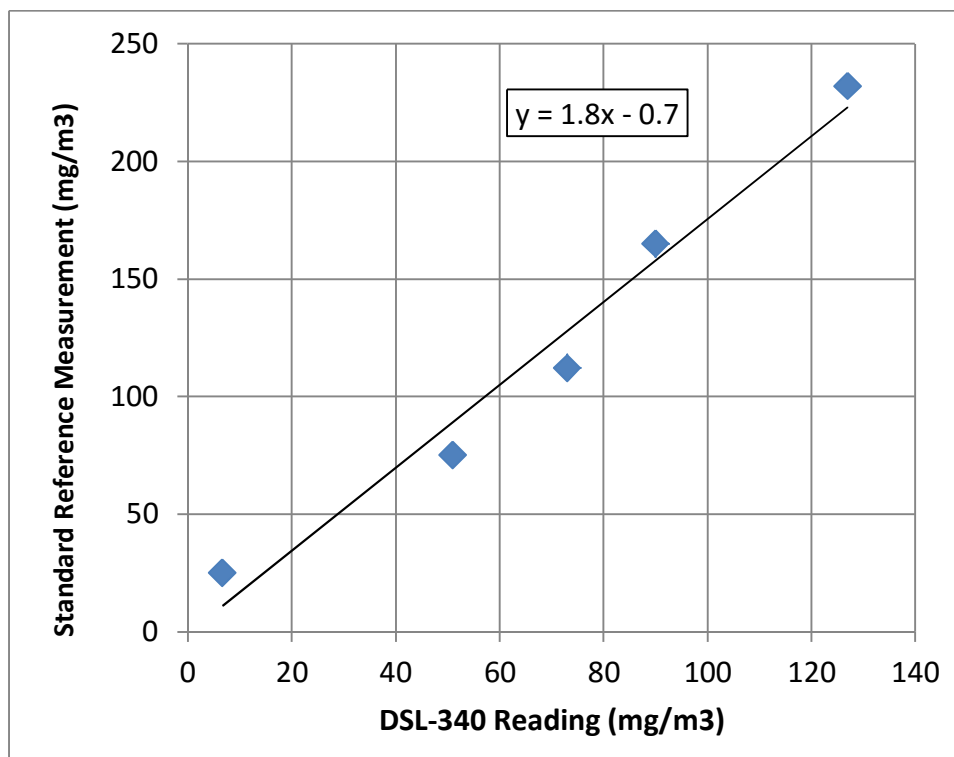


Figure 1: Example of Best Fit Linear Analysis
(DDP Zero=-0.7, DDP Span=1.8)



Warning: After the instrument has been fully calibrated do not adjust the instrument gain. The calibrated DDP Zero value is dependent on the gain. If the gain is adjusted then the DDP Zero calibration must be repeated.

Configuring the outputs

For information on setting up the outputs, refer to the 'Outputs Tab' section.

Configuring the Automatic Calibration Check

The instrument contains a built in electronic calibration check procedure which can be used to confirm that the instrument is still operating within its design parameters. During an automatic calibration check the instrument first

performs a zero check and then a span check. The whole procedure takes about 95 seconds during which the data invalid flag will be enabled.

For the zero check the instrument turns off the LED source and measures the no light condition displaying the resulting zero drift as the measured zero level particle density (mg/m³).

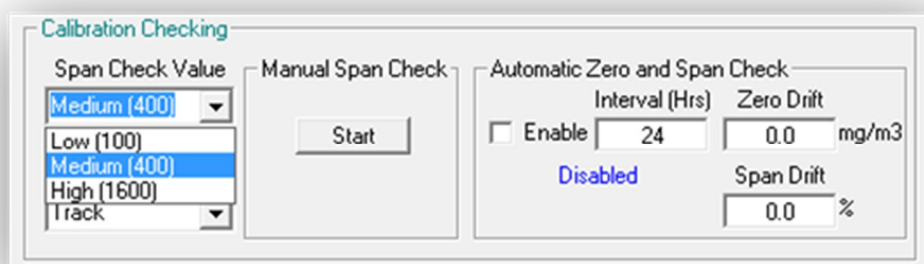
For the span check the instrument produces a controlled intensity fluctuation on the LED light to simulate a high level of particulate or dust. The effect of actual particulates in the stack during the span measurement will add to the reading and so it is removed. The resulting measured span is compared to the span level the instrument expects for the given level of intensity fluctuation and any difference is reported as a percentage drift.

These calibration checks can be operated whilst the instrument is measuring particulates in the stack, and so there is no need to ensure clear path conditions. However, please note that if the stack reading is very high during a calibration check this can introduce larger than normal span drift readings.

To set the instrument up for automatic calibration checking the first step is to set the required span level. Three levels of intensity fluctuation are available (Low, Medium and High). The corresponding span reading depend on the path length and DDP Span settings. For the default settings of 1m path and DDP Span=1 the levels are:

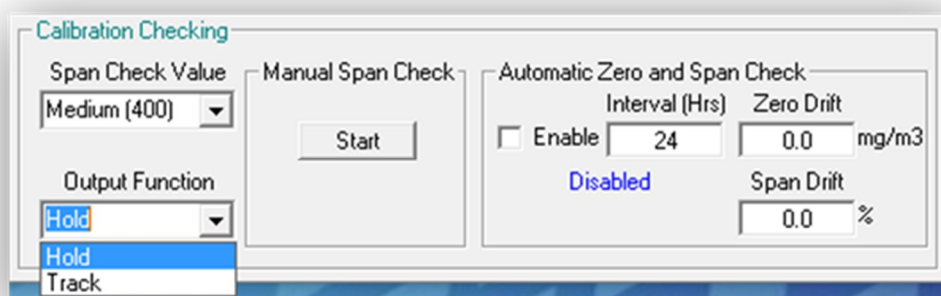
Low = 100mg/m³
 Medium = 400mg/m³
 High = 1600mg/m³

A span level that is close to the maximum measurement range should be selected. The span level is set using the "Span Check Value" drop down menu in the Settings page of the Utility.



During the automatic zero and span check the instrument will flag a fault (Data Invalid) and so any control PC (e.g. SCADA) will know to ignore the readings. However, it is also possible to set the instrument readings to freeze (or Hold) during the zero and span check. This will hold the reading at the value measured just before the start of the automatic zero and span check

routine. This feature is set by using the “Output Functions” drop down menu and selecting “Hold”.



Before enabling the automatic zero and span checking routine it is necessary to set the repeat interval for the automatic checking. This is manually set in the “Intervals” box and can be anywhere from 1 to 1000 hours. The default setting is 24 hours. Note that this time interval can drift by up to ± 10 min/year. This time interval can also be read and manually set in the OI using the **cchr** parameter.

The automatic zero and span calibration is now configured. To enable the function either select the “Enable” box on the settings page of the Utility or set the OI parameter **ccEn** to “yES”.

The first calibration check will occur as soon as the function has been enabled. If it is required for the automatic zero and span check to happen at a particular time of day then the function must be enabled at the correct time. The timing of the automatic zero and span check is controlled within the instrument and should drift by less than ± 10 minutes per year. If at any time the power to the instrument is turned off the timer will restart from the beginning when power is restored, therefore changing the time of day at which the automatic zero and span check will take place.

Each automatic check takes about 95 seconds. During the routine the instrument reading will either track the zero and span checking or will be frozen at the last measured reading (user selected). The data invalid will be triggered producing the faults “ZERO CHECK ACTIVE” and then “SPAN CHECK ACTIVE”.

When the routine has finished the settings page of the utility will display a countdown timer showing when the next automatic zero and span check will take place. Note if the instrument is power cycled this countdown timer will be reset to maximum when the power is restored.

Automatic Zero and Span Check		
	Interval (Hrs)	Zero Drift
☒ Enable	24	0.1 mg/m ³
Enabled: Next 23:58:03		Span Drift
		-0.4 %

After each automatic zero and span check the measured zero and span drift readings are updated and can be read using the Utility or the OI.

Typical zero drifts will be <2mg/m³ and span drifts will be <10%. However, under some conditions they may be greater than these values. One cause of a high apparent span drift is if there is a very high level of particulate in the stack at the time of the span check. The span check routine compensates for the actual stack reading, but this is not perfect and large errors can occur when the actual stack reading is very high and fluctuating.

Maintenance



Caution; care must
be taken

Warning: When fitting or removing the heads, even for routine maintenance, there is a risk that stack gas may escape from the mounting flange, or air-purge body. Stack gas, can be hot, corrosive or otherwise damaging to health. All necessary precautions must be taken to prevent injury being caused by escaping stack gas.



Caution; care must
be taken

Warning: After the instrument has been fully calibrated do not adjust the instrument gain. The calibrated DDP Zero and value is dependent on the gain. If the gain needs to be adjusted then the DDP Zero calibration must be repeated.

The DSL-340 requires very little maintenance because it has no moving parts, no perishable components and is intolerant to changes in the optical signal level.

Checking the Signal Level

The main maintenance consideration is to check the measured signal level. This can drop from the ideal setup value of 4.0V for several reasons, such as dirt build up on the optics, movement of the TRX head, blockages in the flange tubes, or reductions in the brightness of the LED. To a first order changes in the measured signal level have negligible effect on the measured DDP reading. However, if this signal drops by more than 50% from the setup level (e.g. <2.0V) then excess noise and measurement drift can start to degrade the instrument performance.

It is recommended that the measured signal level should be periodically checked either using the **Sig** parameter in the OI or the “Measure Signal” on the settings page of the Utility, to confirm that it is still above 2.0V. If the signal level has dropped do not perform an AGC as this can affect the calibrated DDP Zero settings.

If the measured signal level has dropped below 2.0V and the reference signal is still high then this suggests that either the optical surfaces are dirty or the TRX head alignment has drifted. Either way access to the heads will be required to solve the problem.

Cleaning Optical Surfaces

In order to clean the optical surfaces, first remove the TRX Head from the Air-Purge body by releasing the two latches. Inspect the lenses and remove any heavy dirt deposition with a blast of clean dry air from a clean air canister. Remove any stubborn particles or grit with a camera lens brush. Repeat the process for the Reflector Head.

Once the heavy particles and grit have been removed the optical surfaces can be wiped over with a spectacle cleaning solution and dried with a clean lint free cloth. It is essential that all grit has been removed before wiping over with any cloth. If the lenses are wiped over whilst grit is still present the lenses will be scratched and a permanent signal losses will be introduced.

Checking the Reference Signal

The LED light source in the TRX head is a long life component that should maintain normal operation for at least 5 years, however eventually its light output is likely to diminish to a level where it should be replaced.

The light output of the LED is continuously monitored by an independent photocell located alongside the LED in the TRX head. This photocell generates a signal that is displayed in volts in the reference signal parameter on the Settings tab of the Utility Software or in the rEF parameter in the OI. As the light output falls, so does the reference signal.

The DSL-340 monitors the Reference Signal and will activate the fault alarm when this signal falls below a 1.5V. Ideally operators would also monitor the Reference Signal and consider replacing the TRX head if the signal consistently fell below 2V.

It is therefore recommended that the Reference Signal is checked on a 6 month interval.

Fault Messages and Trouble Shooting

Fault and Warning messages can appear either on the Readings tab of the Utility software, or on the display of the OI.

Important fault or warning messages that appear in the Utility Software will remain permanently displayed on the Readings tab until the fault or warning has been cleared and the operator has clicked on the message. This means that even if a fault occurs only temporarily and has been resolved before the operator views the Utility Software, the fault message will remain visible until the operator has clicked on the text to acknowledge it. Fault message text can only be cleared if the fault has been resolved.

Where messages appear on the OI they will flash intermittently on the display. This means that the operator can still navigate and use the OI, but will be constantly reminded of the fault by the flashing message. Flashing messages will automatically disappear when the fault is resolved.

Fault messages indicate there is a fault with the instrument and that the readings are not valid. Warning messages indicate that there may be a problem and the warning should be investigated.

The DSL-340 TRX Head is the intelligent centre of the instrument, and all measurements, calculations, and data storage is done in the TRX Head. The OI is a terminal which presents data from the TRX and provides an interface through which the operator can alter settings and parameters in the TRX Head.

The Utility software runs on a PC or laptop which can be connected via USB to either the DSL-340 TRX Head, or to the OI. If a PC is connected to the OI, instead of the TRX, the OI simply becomes a conduit through which data is passed between the PC and the TRX. In this scenario, if there is a fault with the OI, or if the OI loses communications with the TRX, or if the OI loses power, the Utility software will be unable to retrieve the required data from the TRX, and the Utility Software will disconnect.

The following section describes each fault and warning message, outlines potential causes and highlights actions that can be taken to try and resolve the issue. If the actions described do not solve the problem then further advice should be sought from your DynOptic Systems representative.

Fault Messages

LO REF-VIS (“Lo r”)

This fault indicates that the Reference Signal has fallen below 1.5V. The Reference Signal is a measure of the LED brightness, made by a photocell within the TRX head. This error is usually caused by a dim (or failed) LED in the TRX head.

However, it can also be triggered by a failure of the LED drive circuit, a low power supply voltage to the DSL-340 (should be 24VDC), or a failure of the Reference Signal measurement circuit.

Check the 24Vdc power supply to the TRX head and check the Reference Signal value on the Settings tab in the Utility Software. If 24Vdc is correct and the reference signal is <1.5V then there is a problem with the circuit board or LED within the TRX head. The TRX head should be swapped out and returned to DynOptic Systems for repair.

In the event that the TRX is swapped out it will be necessary to complete the full setup, calibration and commissioning process, as described in the earlier section in this manual, using the replacement TRX head.

HI REF-VIS (“Hi r”)

This fault indicates that the Reference Signal has risen above 4.9V. The Reference Signal is a measure of the LED brightness, made by a photocell within the TRX head. The DSL-340 electronics can only measure a maximum signal of 5.0V, so any voltages above this level will saturate the measurement circuit.

This fault can be caused by a failure of the reference signal measurement circuit, or by increased brightness of the LED.

In some instances it may be possible to adjust the Reference Signal back to 4.0V using the potentiometer marked VR1 on the TRX circuit board. This potentiometer is used to adjust the Reference Signal voltage during test at our factory. After adjustment it is sealed with a red varnish. Before attempting such adjustments you should contact DynOptic Systems for further advice, which will vary depending on the age of the instrument and other faults that may be present.

HI MEAS-VIS (“Hi S”)

This fault indicates that the Measure Signal has risen above 4.9V. The Measure Signal is the main signal used to generate the instruments measurement reading. The Measure Signal is a voltage generated by light returned from the Reflector head falling on the main photocell in the TRX Head.

The DSL-340 electronics can only measure a maximum signal of 5.0V, so any signal above this voltage will cause saturation, so this fault message serves to warn of saturation in the measurement circuit.

Saturation of the measurement circuits will frequently occur during the installation, calibration and commissioning phase and will normally be resolved during the alignment and AGC process.

If this fault occurs during normal operation it may indicate that the original alignment or setup was not performed correctly or there has been an electronics failure.

If the message appears during normal operation the following actions should be considered in order:

- Repeat the setup, calibration and commissioning procedure (i.e. AGC and DDP Zero adjust)
- Seek further assistance

OVERRIDE ACTIVE

During set-up and commissioning of the instrument it is possible to manually set the DSL-340 into reading override mode. This mode is used to confirm the correct operation of the SCADA or data logging system. When in this mode the fault relay is triggered and this fault message is displayed. The message disappears when the unit returns to normal reading mode.

If this message appears during normal operation then the instrument has been accidentally switched into reading override mode. It should turn itself off after 5 minutes.

“HEAd” (only displayed on the OI)

This message is specific to the OI. This message indicates that the OI has lost communications with the DSL-340 TRX Head.

This fault is usually caused by a problem with the cable between the OI and the TRX Head. Check that the wiring connections have been made in the correct order (consult the Installation Manual), and that all the connections are sound. Ensure that all two part terminals have been pushed in properly and in

the correct orientation. Check that the cable is intact and has not been severed or damaged.

This problem can also be caused by having an incorrect combination of negative and positive bias applied to the RS485 signals in the interconnecting cable, or by missing termination links. Check that the RS485 bias links and termination link (marked as POS BIAS, NEG BIAS, and TERMINATION) are properly fitted on the main OI circuit board (up underneath the front panel). Also check that the termination link (marked as LAST HEAD) is fitted on the circuit board in the TRX Head.

This message will also appear if the TRX Head loses power. Check that the TRX has power (use a DVM or open the TRX Head and look for flashing LEDs on the TRX circuit board).



Warning: When a HEAD fault is activated the OI freezes all readings, values, and settings at the last known levels. This includes the measurement reading, analogue output levels, and Level Alarm status, with the exception of the Service Alarm which will activate to indicate the fault.

SPAN CHECK ACTIVE (SP.cc)

This fault message is displayed only when the instrument is performing a span check.

ZERO CHECK ACTIVE (Zr.cc)

This fault message is displayed only when the instrument is performing a zero check.

Warning Messages

ALARM ACTIVE

This warning indicates that an alarm threshold has been passed.

LO MEAS-VIS (“Lo S”)

This fault indicates that the Opacity reading measured by the instrument, has fallen below 50%. DDP works independently of Opacity, but if Opacity has fallen below 50% it indicates that the DDP signal is likely to be degraded.

Usually this message is caused by dirty optics, a misaligned TRX head or something blocking the beam. The first action should be to clean the windows and check the head alignment.

If this does not solve the problem then the instrument could be setup again from the beginning, including a new AGC and DDP Zero setting.

MEMORY FAULT (“EEch”)

The DSL-340 employs CRC checking on the EEPROM memory. Each time the instrument is powered up the CRC checking compares the state of EEPROM that it last knew, against the state of the EEPROM memory on power up.

If the instrument finds a discrepancy between the two EEPROM memory states, a memory fault is registered.

On registering the memory fault the instrument performs a self-checking routine which analyses each byte of memory and ensures that the values stored there are within an acceptable range. Values outside the acceptable range are considered to be corrupted values and the memory byte is replaced with a default value from FLASH memory. Essentially the instrument heals itself.

Memory corruption is rare, but can occur in certain circumstances, for example if the instrument suffers a power loss whilst writing to its EEPROM memory.

Discrepancies can also be caused by a software upgrade (in the TRX head or the OI, not the Utility software), so they can be an expected result of such actions.

On encountering a memory fault operators should always power cycle the instrument, i.e. switch it off, then on again. This action should clear the fault because self-healing process will have installed a new value in the corrupted memory byte, and the compared states of the power off memory and the power on memory should now match.

If a power cycle does not clear the fault it is likely that there is a more serious problem and you should contact DynOptic Systems or your distributor for further assistance.

Note: *that when the instrument self-heals by replacing corrupted data with default data from FLASH memory, some operational settings and values may have been overwritten. It is therefore essential that all parameters, settings, and values are checked for validity by the operator immediately after resolving a MEMORY fault.*

COMMS CRC FAULT (“CrC”)

When the TRX head is connected to an OI it sends a check signal (CRC) to test the quality of the communication line. This error message appears if the communication check fails.

This message may indicate one of the following:-

- The communications line is too noisy, this may be due to the close proximity of high power or AC voltage cables.
- There is a problem with the set-up of the TRX to OI communication cable (e.g. poor grounding, bad connection).

© All content copyright of DynOptic Systems Ltd 2016.